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CSA 6101
Digital forensics and cybercrime investigation

CO2 - AT3

⇒ Phishing Email Investigation:

→ Identifactions:

→ verify sender's email address and domain.
→ check suspicious links and attachments.
→ Examine email headers for spoofing.
→ Identify urgent or misleading language.

→ Evidence collections:

→ Save the phishing email.
→ collect email headers, attachments & URLs.
→ obtain firewall, email server and login logs.
→ Record affected user accounts.

→ Evidence Preservation:

→ Create forensic copies of emails.
→ Generate SHA-256 hash values.
→ Store evidence in read-only storage.
→ maintain a chain of custody.

→ Analysis:

→ Trace the Sender's IP address.
→ Determine how 18 accounts were compromised.
→ Analyze phishing indicators and malware.
→ Assess the impact on organization data and systems.

⇒ Ransomware Investigation:

→ Identification:

- → Identification the infected us computers.
- → Detect encrypted files and ransom notes
- → Confirm the 2TB data encryption and 6-hour Service disruption

→ Evidence Collection:

- → collect windows event logs, firewall logs, antivirus logs, memory dump and ransom notes
- → Acquire forensic disk images and network traffic captures.

→ Evidence preservation:

- → Isolate infected system.
- → Create forensic image.
- → Generate SHA-256 hashes.
- → preserve evidence using chain of custody.

→ Analysis:

- → Identify the ransomware family.
- → Determine the infection source.
- → Analyze attack timeline and affected systems
- → Evaluate business impact and recovery options.

→ Reporting:

→ Document findings, evidence, impact and recommendations.

→ Suggest back up recovery, path managment, MFA and user awareness training.

⇒ Network forensic Investigation:

→ Identification:

→ Investigation 12 unauthorized login attempts

→ Analyze 8 suspicious IP addresses.

→ Examine the 25GB abnormal outbound traffic.

→ Evidence collection:

→ collect firewall logs, IDS/IPS logs, VPN logs, system logs and packet captures (PCAP).

→ Record login histories and network traffic.

→ Evidence Preservation:

→ create forensic copies of logs.

→ Generate SHA-256 hashes.

→ Store evidence securely.

→ maintain a Chain of custody.

→ Analysis:

→ Identify the attack technique.

→ Trace attacker IP addresses.

→ Analyze login attempts and data exfiltration.

→ Determine affected system and impact.

→ Investigation Report :

Summarize the incident, evidence collected, attack timeline, findings, impact and recommendations for preventing future attacks.

⇒ conclusion :

effective identification, evidence handling, analysis and reporting helped Investigation the incidents, minimize security risks, and strength the organization's cybersecurity posture.